

# certo-vision: calibrated typed image decisions from a few dozen labelled images

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## Abstract

Many product decisions about an image are small and typed: *is this part defective?*, *which of these four defect types is it?*, *how ripe is this fruit on a four-level scale?* They need a calibrated probability, a separate statement of whether the model is in its depth, and a refusal when it is not. Generative vision-language models answer such questions but do not give those three things cheaply. We describe certo-vision, a small decision layer over a frozen SigLIP 2 image-text encoder that answers `noul` (yes/no), choice and score questions in one embedding pass, fits a per-question temperature and per-option bias on a few dozen labelled images, blends text prototypes with image prototypes, and gates every answer with a nearest-neighbour scope check set to keep 95% of in-scope images. On nine typed questions drawn from two public datasets that resemble client work (MVTec AD industrial inspection, banana ripeness), the few-shot-calibrated decider reaches 93–95% on gross yes/no defect questions, 0.87 on a four-way defect type, and a mean absolute error of 0.23 levels on a four-level ripeness score, while rejecting 100% of out-of-scope images, including other factory parts photographed in the same style. It fails on subtle defects of a small part (screw: 0.71 yes/no, chance on defect type), which locates the ceiling of pooled embeddings. Zero-shot text prototypes alone are at or near the majority rate on the industrial questions, and two popular confidence signals (max probability, truncated- dimension agreement) do not detect out-of-scope input. We release the code, the calibration recipe, a demo calibration, and the evaluation harness.

## 1. The contract

A typed decision API (as popularised by Jev from TypeSafe AI for text) takes one *state* and a schema of questions, and returns for each question a value of the declared type with calibrated probabilities. The caller never parses free text. We keep that contract for images and add two fields the caller can act on:

- **confidence**: how close the image is to the images the question was calibrated on (a scope score, not the answer probability);
- **abstain**: whether confidence is below a threshold fitted to keep 95% of in-scope images.

The design constraint is cost. Every question is answered from the same embedding of the image, so a schema of fifty questions costs one encoder pass plus fifty dot products. That is what makes it viable to put ten or twenty typed checks on every frame or every upload, which a generative model call per question is not.

## 2. Method

**Backbone.** `google/siglip2-so400m-patch14-384` (about 1.1 B parameters, 1152-d joint image-text space), frozen. An earlier prototype used `EmbeddingGemma2`; Section 4.1 reports the comparison that settled the choice.

**Prototypes.** Each option of a question is rendered as a short text (“Is the banana overripe? — yes, the banana is overripe with brown spots or mostly brown”) and embedded once. Probabilities are a softmax over the dot products between the unit image embedding and the unit option prototypes.

**Calibration.** On a labelled calibration set we fit one temperature and one bias per option by minimising negative log-likelihood (plain gradient descent on a problem with at most 256 parameters). This is the stand-in for the reinforcement-learned calibration a text decision model gets at training time; here it is cheap enough to run per deployment.

**Few-shot prototypes.** With the same labelled images, each text prototype is blended with the mean embedding of that option’s images,  $p \leftarrow \text{norm}(0.7 \cdot p_{\text{text}} + 0.3 \cdot \text{mean\_img})$ , the Tip-Adapter idea reduced to one line. This is the step that carries almost all the gain (Section 4.2).

**Scope gate.** The calibration embeddings are kept. For a new image, confidence is its highest cosine similarity to any calibration image, divided by the median leave-one-out nearest-neighbour similarity within the calibration set, clipped to  $[0, 1]$ . The abstain threshold is the 5th percentile of that quantity over the calibration set itself, so the gate is set to keep 95% of images like the ones it was calibrated on. Nothing about the gate depends on the question’s labels.

**Score questions** return the expected level under the calibrated distribution, with the legend.

**Persistence.** Prototypes, temperature, bias, calibration embeddings and the threshold are saved to one file and loaded by the server. A served request matches a saved question by its id and its exact spec, so a schema edit silently drops back to zero-shot and is reported as `calibrated: false`.

### 3. Protocol

- Every image set is split 30% calibration / 70% test, redrawn for each of five seeds; we report mean  $\pm$  standard deviation over seeds and always the majority-class rate of the test split.
- Three variants per question: zero-shot (text prototypes, default temperature), calibrated (fit on the 30%), calibrated + few-shot (fit and blend on the same 30%).
- Metrics: accuracy, negative log-likelihood, Brier score, expected calibration error (15 bins), area under the risk-coverage curve; for score questions also mean absolute error in levels and Spearman  $\rho$ .
- Scope: the gate is fitted on the seed-0 calibration split; we report the in-scope abstain rate on the test split, and on each out-of-scope set the AUROC of in-scope vs out-of-scope confidence and the fraction rejected at the fitted threshold.
- Data. STL-10 (10 classes, 100 per class, 96 px) for the backbone comparison, with DTD textures, EuroSAT satellite tiles and Oxford-IIIT pet photos (200 each) as out-of-scope sets. MVTec AD bottle, hazelnut, carpet and screw (100 good training images plus the full test split per object) and the BananalmaImageBD ripeness set (150 per level, four levels) for the trial. Out-of-scope sets for the trial are the *good* images of the other three MVTec objects, DTD and STL-10 cats.
- Hardware. One NVIDIA GB10 (DGX Spark). Embeddings are computed once and cached; everything else is numpy.

## 4. Results

### 4.1 Backbone

| Backbone         | task                      | zero-shot | calibrated | few-shot                 | majority | ECE few-shot | embed ms (GB10) |
|------------------|---------------------------|-----------|------------|--------------------------|----------|--------------|-----------------|
| SigLIP 2 so400m  | subject (10-way)          | 0.990     | 0.993      | <b>0.995</b> $\pm$ 0.001 | 0.110    | 0.005        | 72              |
| SigLIP 2 so400m  | is it a cat?              | 0.900     | 0.900      | <b>0.990</b> $\pm$ 0.005 | 0.900    | 0.016        | 72              |
| SigLIP 2 so400m  | blur level (5, synthetic) | 0.247     | 0.340      | <b>0.878</b> $\pm$ 0.004 | 0.207    | 0.028        | 72              |
| EmbeddingGemma 2 | subject (10-way)          | 0.967     | 0.983      | <b>0.988</b> $\pm$ 0.002 | 0.110    | 0.011        | 84              |
| EmbeddingGemma 2 | is it a cat?              | 0.126     | 0.900      | <b>0.964</b> $\pm$ 0.021 | 0.900    | 0.033        | 84              |
| EmbeddingGemma 2 | blur level (5, synthetic) | 0.232     | 0.202      | <b>0.857</b> $\pm$ 0.015 | 0.207    | 0.043        | 84              |

Table 1. STL-10, 1000 images, five seeds. SigLIP 2 is better on every task and variant, is faster, and carries no audio weights. The zero-shot `nou_l` on EmbeddingGemma 2 (0.126) answers “yes” to nearly everything: the negated prototype does not work there.

Two things are visible already. A calibrated `nou_l` without image prototypes equals the majority rate exactly (0.900 on both backbones): temperature and bias cannot add information, they only reshape the prior. And the synthetic blur score goes from chance to 0.88 only once image prototypes are blended in.

## 4.2 Trial on public stand-ins for client questions

| Question                          | type   | n   | calib imgs /<br>option<br>(min) | zero-shot | calibrated | few-shot                    | majority | ECE<br>few-shot | MAE / $\rho$ | in-scope<br>abstain | OOD<br>rejected (n) |
|-----------------------------------|--------|-----|---------------------------------|-----------|------------|-----------------------------|----------|-----------------|--------------|---------------------|---------------------|
| bottle:<br>defective?             | noul   | 183 | 19                              | 0.640     | 0.874      | <b>0.932</b> $\pm$<br>0.011 | 0.640    | 0.056           |              | 0.10                | 100% (659)          |
| hazelnut:<br>defective?           | noul   | 210 | 21                              | 0.751     | 0.950      | <b>0.951</b> $\pm$<br>0.006 | 0.658    | 0.033           |              | 0.06                | 100% (639)          |
| carpet:<br>defective?             | noul   | 217 | 27                              | 0.600     | 0.641      | <b>0.940</b> $\pm$<br>0.019 | 0.578    | 0.051           |              | 0.05                | 100% (651)          |
| screw:<br>defective?              | noul   | 260 | 36                              | 0.537     | 0.617      | <b>0.712</b> $\pm$<br>0.024 | 0.537    | 0.095           |              | 0.04                | 100% (638)          |
| hazelnut:<br>defect type<br>(4)   | choice | 70  | 5                               | 0.490     | 0.837      | <b>0.869</b> $\pm$<br>0.027 | 0.274    | 0.098           |              | 0.10                | 100% (639)          |
| screw:<br>defect type<br>(5)      | choice | 119 | 7                               | 0.243     | 0.200      | <b>0.364</b> $\pm$<br>0.054 | 0.238    | 0.131           |              | 0.02                | 100% (638)          |
| banana:<br>ripeness<br>(score, 4) | score  | 600 | 45                              | 0.483     | 0.626      | <b>0.833</b> $\pm$<br>0.025 | 0.267    | 0.044           | 0.23 / 0.90  | 0.09                | 100% (370)          |
| banana:<br>overripe?              | noul   | 600 | 45                              | 0.799     | 0.801      | <b>0.929</b> $\pm$<br>0.014 | 0.745    | 0.028           |              | 0.09                | 100% (370)          |
| banana:<br>stage (4)              | choice | 600 | 45                              | 0.499     | 0.600      | <b>0.785</b> $\pm$<br>0.028 | 0.267    | 0.051           |              | 0.09                | 100% (370)          |

Table 2. SigLIP 2, five seeds, 30/70 split. “OOD rejected” is the lowest rejection rate over that recipe’s out-of-scope sets at the threshold fitted for 95% in-scope coverage; n is the total number of out-of-scope images.

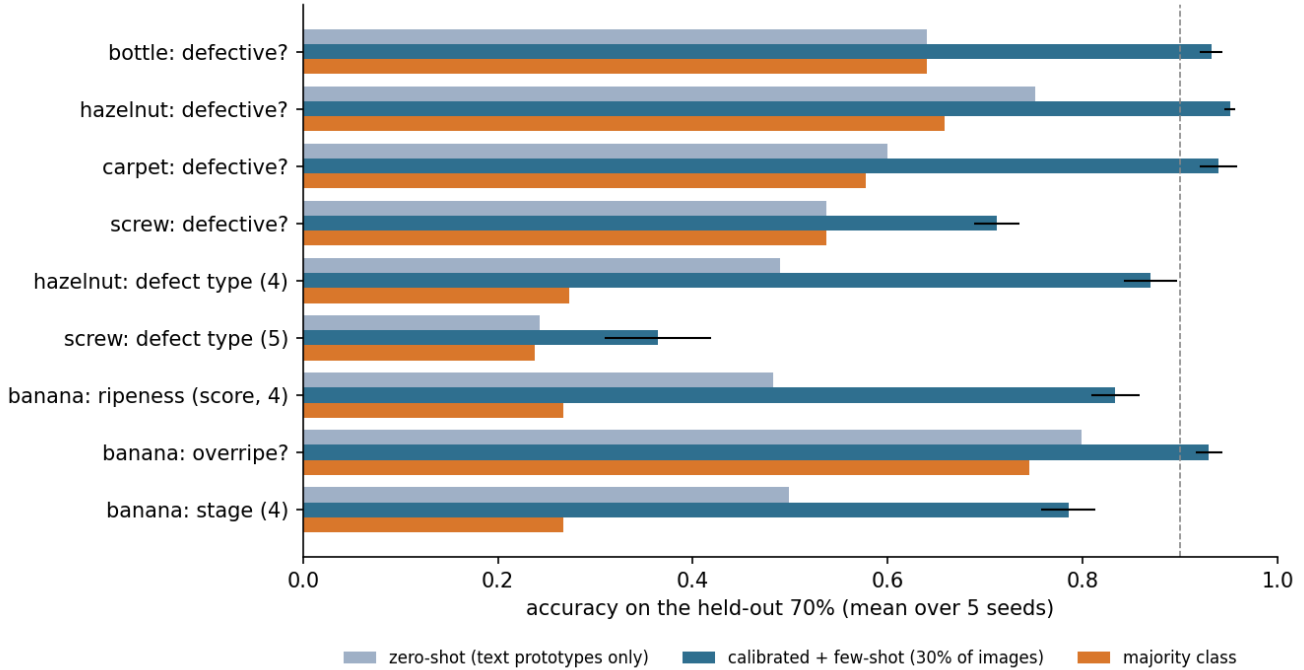


Figure 1: Accuracy per question: zero-shot, calibrated + few-shot (error bars: std over five seeds), and the majority class. The dashed line is 0.9.

**Coarse decisions pass.** Gross defects on bottle, hazelnut and carpet and the two banana yes/no or score questions land between 0.93 and 0.95 with 19–45 calibration images per option. The ripeness score, the first test of the score type on real ordinal labels, is within a quarter of a level on average ( $\rho = 0.90$ ). ECE after calibration is 0.03–0.06 on these rows.

**Fine-grained decisions fail.** The screw set is thread damage, scratches and tip chips on a small object against a plain background. A pooled 1152-d embedding of the whole image does not retain that detail: 0.71 on yes/no

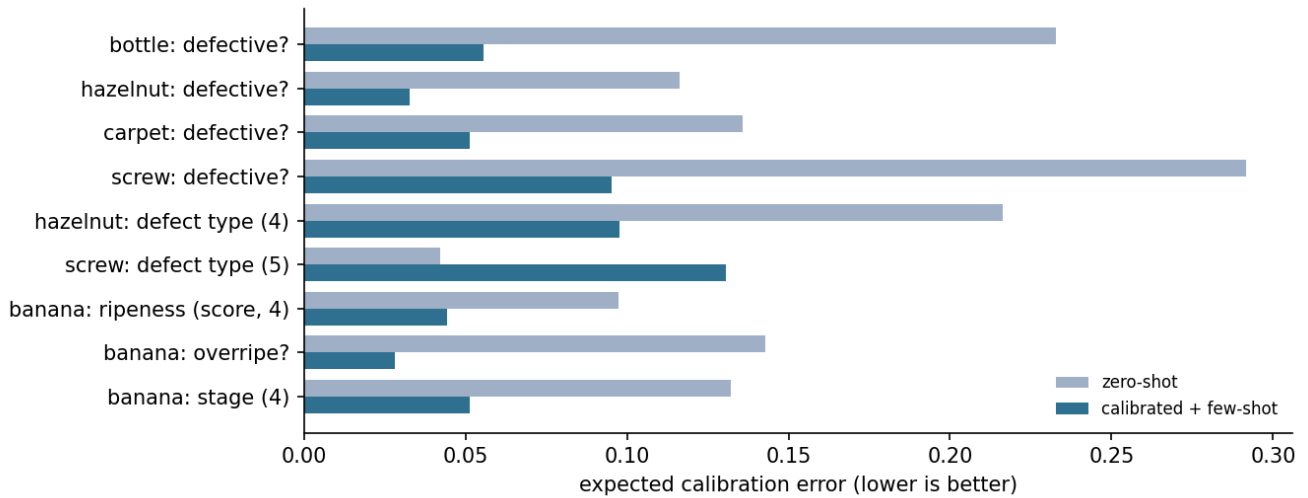


Figure 2: Expected calibration error, zero-shot vs calibrated + few-shot. Calibration brings every question under 0.1 and most under 0.06.

and chance on type, with the worst calibration of the table. Patch-level methods on MVTec (WinCLIP, PatchCore) solve exactly this, and that is the path if such questions are in scope.

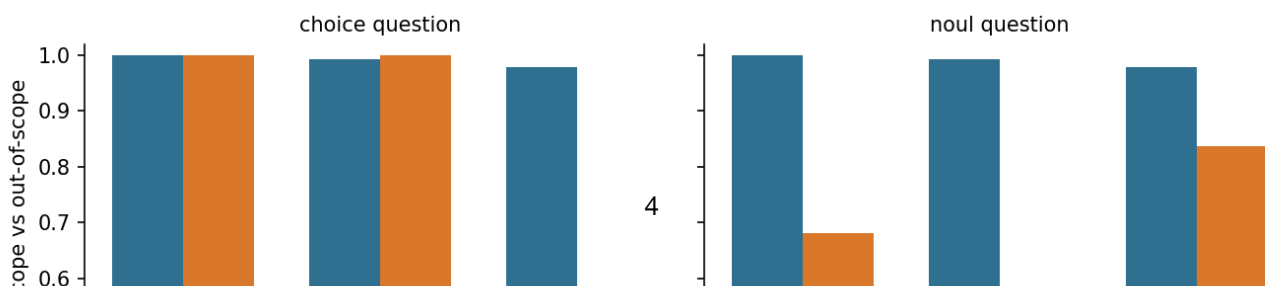
**Zero-shot is not a product.** On the industrial yes/no questions it is at or within two points of the majority rate, and on the banana questions it is 25–35 points under few-shot. Its ECE is 0.2–0.4 everywhere. The labelled images are the product; the text prototypes are a prior that the images refine.

**Thin options are noisy.** The defect-type rows had five to seven calibration images per option, and their seed spreads run to 5 points. Treat those rows as indicative; the calibration tool warns below ten images per option.

#### 4.3 Scope: what the gate detects

| Backbone         | question | out-of-scope set    | AUROC NN ratio | AUROC max-prob | AUROC MRL agreement | rejected by NN gate | max-prob > 0.9 |
|------------------|----------|---------------------|----------------|----------------|---------------------|---------------------|----------------|
| SigLIP 2         | choice   | textures (DTD)      | 1.000          | 0.999          | 0.722               | 100%                | 1%             |
| SigLIP 2         | choice   | satellite (EuroSAT) | 0.992          | 1.000          | 0.669               | 100%                | 0%             |
| SigLIP 2         | choice   | Oxford pets         | 0.979          | 0.567          | 0.352               | 93%                 | 100%           |
| SigLIP 2         | noul     | textures (DTD)      | 1.000          | 0.680          | 0.625               | 100%                | 97%            |
| SigLIP 2         | noul     | satellite (EuroSAT) | 0.992          | 0.484          | 0.600               | 100%                | 100%           |
| SigLIP 2         | noul     | Oxford pets         | 0.979          | 0.837          | 0.882               | 93%                 | 79%            |
| EmbeddingGemma 2 | choice   | textures (DTD)      | 1.000          | 0.993          | 0.923               | 100%                | 2%             |
| EmbeddingGemma 2 | choice   | satellite (EuroSAT) | 1.000          | 0.926          | 0.917               | 100%                | 40%            |
| EmbeddingGemma 2 | choice   | Oxford pets         | 0.996          | 0.731          | 0.625               | 98%                 | 100%           |
| EmbeddingGemma 2 | noul     | textures (DTD)      | 1.000          | 0.424          | 0.408               | 100%                | 96%            |
| EmbeddingGemma 2 | noul     | satellite (EuroSAT) | 1.000          | 0.356          | 0.371               | 100%                | 100%           |
| EmbeddingGemma 2 | noul     | Oxford pets         | 0.996          | 0.729          | 0.744               | 98%                 | 58%            |

Table 3. STL-10 questions calibrated on 300 images (seed 0); 700 in-scope test images vs 200 per out-of-scope set. “MRL agreement” is the agreement between answers at truncated embedding widths (128 to full), the Matryoshka ensemble idea; for SigLIP 2, which is not Matryoshka-trained, it is included only to show it does not help there either.



The Oxford pets row is the important caveat. Those are cats and dogs, the question is “is it a cat?”, and the model answers them mostly correctly. The gate still rejects 93–98% of them because they are high-resolution photographs and the calibration set is 96-pixel STL-10. **The gate detects domain shift, not off-topic questions.** For deployment that means calibration images must come from the deployment camera, and the in-scope abstain rate must be reported as a first-class metric next to accuracy. The trial in Section 4.2 is the favourable case: the out-of-scope sets include the other MVTec objects, shot in the same style, and the gate still rejected all of them.

#### 4.4 Where the ceiling is: a form with handwritten digits

A worked example in the repository (`examples/forms/`) puts a number on the fine-detail limit. 600 synthetic form snippets carry a printed field name (six options) and four boxes with MNIST digits. Read through the whole image, the field name is 100% after calibration (96% zero-shot), but each digit is only 29–50% on a 10-way question with the worst calibration error in this report, and “is the number  $\geq 5000$ ” is 0.69. Cropping each box first and asking one digit question per crop gives  $0.918 \pm 0.005$  per digit and 0.707 per whole number. Printed words are within scope of the pooled embedding; handwritten digits need a crop step in front of the decider, and beyond about 92% per digit they need a digit classifier.

#### 4.5 Cost

Embedding one image takes 72 ms at batch size one on the GB10 (fp32, no compilation); answering the whole schema afterwards is about 1 ms for two questions and grows linearly in the number of options. The calibration file for the three-question banana demo is a few megabytes and loads in milliseconds.

### 5. What this does and does not show

Shown: a frozen image-text encoder with per-question calibration and few-shot prototypes gives usable, calibrated typed decisions on coarse visual questions from a few dozen labelled images, with a scope gate that reliably rejects images unlike the calibration set.

Not shown:

- **Cross-camera transfer.** Every trial calibrates and tests on the same source. The pets result says the gate will reject a different camera; we have not measured how much accuracy survives when the gate is re-fitted on the new camera. This is the first thing to test on a real client set.
- **Comparison to a VLM.** We did not run a generative vision-language model on the same typed questions. The claim is cost and calibration, not that a VLM would be less accurate.
- **Fine detail.** The screw failure is diagnosed, not fixed, and the forms example shows the same limit on handwritten digits. A trained token-level pooling head or a patch-based fallback is the obvious next experiment and is out of scope for this release.
- **Ordinal scores beyond one dataset.** The blur score is synthetic; the ripeness score is one real set.
- **Thin options.** Defect-type rows had under ten calibration images per option.

### 6. Release scope

What the released tool promises: give it thirty to fifty labelled images per option from the deployment camera, and it returns calibrated typed answers plus an abstain flag, for questions a person could answer from a thumbnail. Questions that need fine detail (small defects, text, counting, spatial relations) are outside that promise and should route to a different model. The calibration command runs a held-out check and prints accuracy, calibration error, majority rate and abstain rate before saving anything, so the scope is checked per deployment, not assumed.

### 7. Related work

Prototype classifiers over CLIP-style encoders and their few-shot refinements (CoOp, Tip-Adapter) are the basis of the prototype and blending steps. Temperature scaling (Guo et al.) is the calibration step, extended with a per-option bias. Nearest-neighbour distance as an out-of-distribution score (Sun et al.) is the gate. That pooled embeddings miss small defects on MVTec AD, and that patch-level matching fixes it, is established by WinCLIP and PatchCore. The typed decision contract with calibrated probabilities and a separate confidence follows Jev

(TypeSafe AI) for text. The contribution here is the combination into one cheap, calibrated, abstaining decision layer, the evaluation protocol with majority baselines and scope metrics, and the negative results on zero-shot prototypes and on max-probability and truncation-agreement as scope signals.

## 8. Reproduction

```
pip install -e ".[siglip,serve,dev]"
python -m pytest tests # stub backbone, no weights
# data export (public sets, into data/)
python tools/make_mvtec.py --out data/mvtec --objects bottle screw hazelnut carpet --good-train 10
python tools/make_banana.py --out data/banana --per-class 150
# trial, five seeds, cached embeddings
certo-vision trial --recipe recipes/banana.json --images data/banana --seeds 0 1 2 3 4
# calibrate for serving, with the held-out check
certo-vision calibrate --recipe recipes/banana.json --images data/banana --out calib.npz
certo-vision serve --calib calib.npz --recipe recipes/banana.json --port 8080
python report/make_figures.py # figures and tables from results/
```

results/ holds the JSON every number in this report was read from.

## References

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